



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Experiment - 06

Vacuum Cleaner Problem

Aim

To implement the Vacuum Cleaner problem using a Simple Reflex Agent.

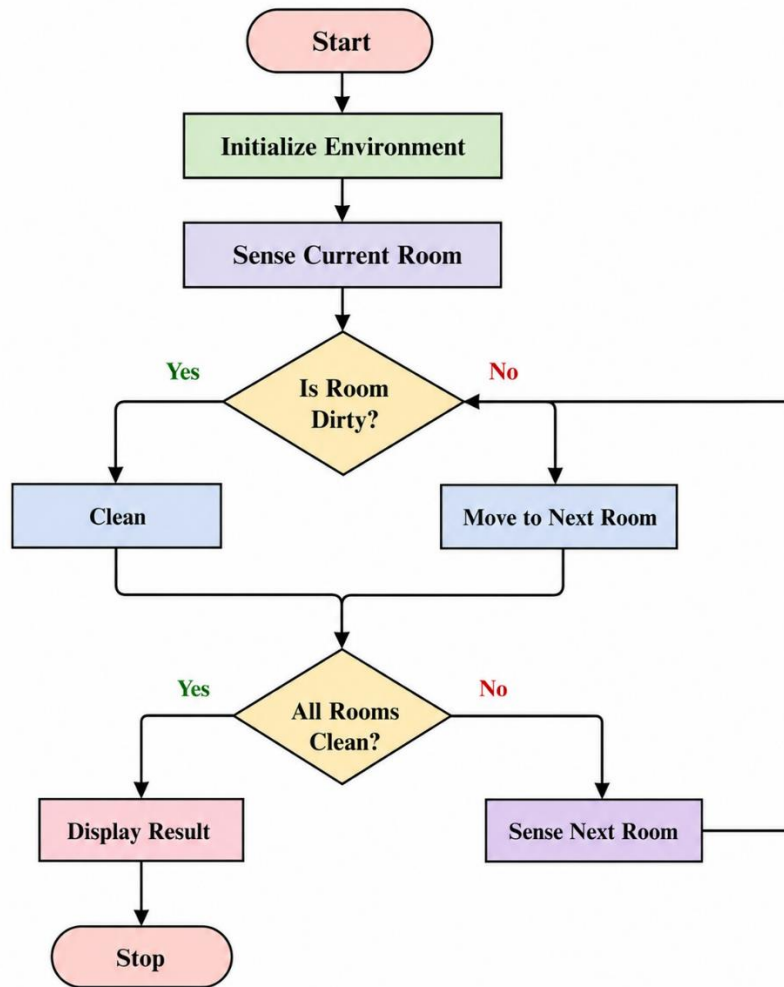
Objective

To clean all dirty rooms by sensing the environment and performing the appropriate action (Clean, Move Left, or Move Right).

Algorithm

1. Start with the vacuum cleaner in the initial room.
2. Sense the current room's status (Clean or Dirty).
3. If the room is dirty, clean it.
4. If the room is clean, move to the next room.
5. Repeat the process until all rooms are clean.
6. Display the actions performed by the vacuum cleaner.
7. Stop.

FlowChart



Code:

```
rooms = {'A': 'Dirty', 'B': 'Dirty'}
```

```
current_room = 'A'
```

```
while True:
```

```
    print("Current Room:", current_room)
```

```
    print("Status:", rooms[current_room])
```

```
    if rooms[current_room] == 'Dirty':
```

```
        print("Action: Clean")
```

```

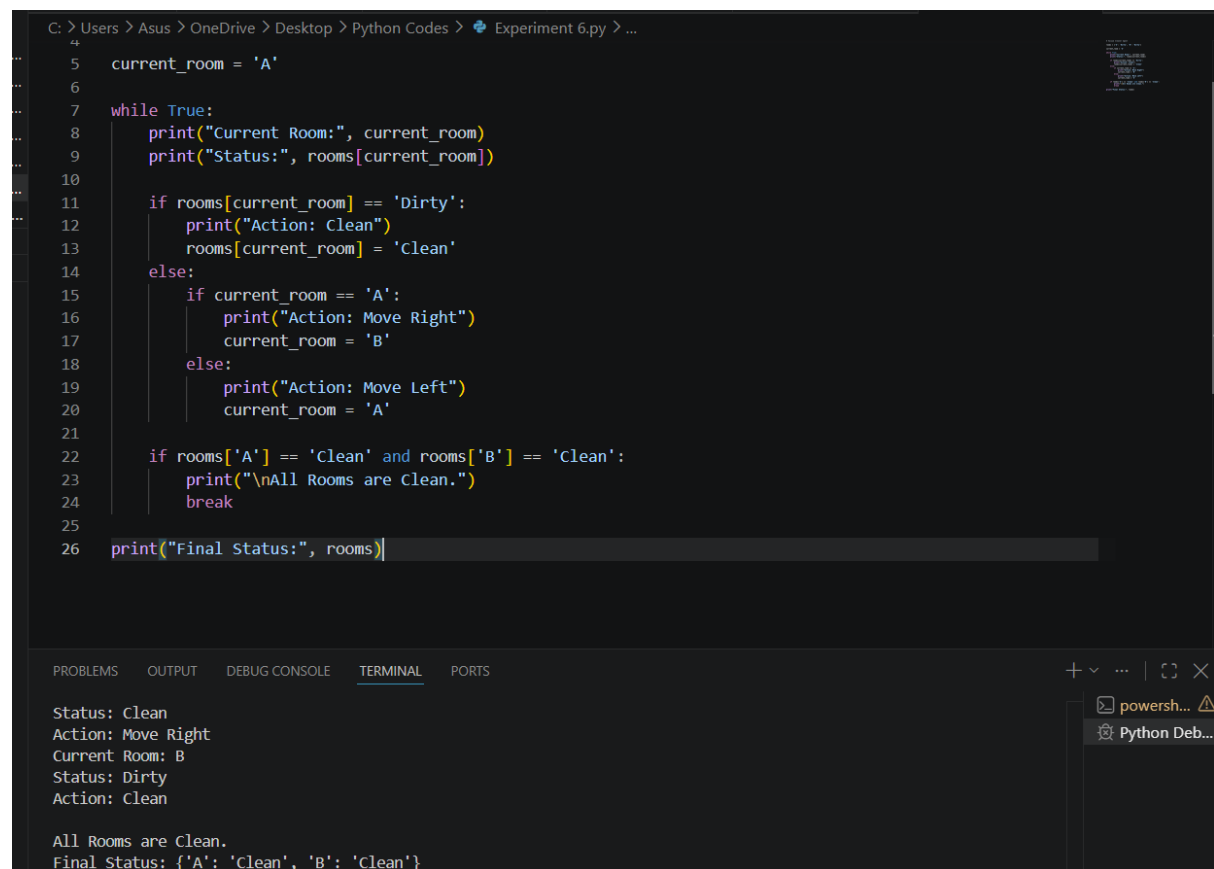
        rooms[current_room] = 'Clean'
else:
    if current_room == 'A':
        print("Action: Move Right")
        current_room = 'B'
    else:
        print("Action: Move Left")
        current_room = 'A'

if rooms['A'] == 'Clean' and rooms['B'] == 'Clean':
    print("\nAll Rooms are Clean.")
    break

print("Final Status:", rooms)

```

Output:



The screenshot shows a VS Code editor window with a Python file named 'Experiment 6.py'. The code is a while loop that checks the status of two rooms, 'A' and 'B'. If a room is 'Dirty', it is cleaned. If a room is 'Clean', the current room is moved to the adjacent room (A to B, B to A). The loop continues until both rooms are 'Clean', at which point it prints the final status and breaks. The terminal output shows the execution of the code, including the status of the rooms and the actions taken.

```

C: > Users > Asus > OneDrive > Desktop > Python Codes > Experiment 6.py > ...
4
5  current_room = 'A'
6
7  while True:
8      print("Current Room:", current_room)
9      print("Status:", rooms[current_room])
10
11     if rooms[current_room] == 'Dirty':
12         print("Action: Clean")
13         rooms[current_room] = 'Clean'
14     else:
15         if current_room == 'A':
16             print("Action: Move Right")
17             current_room = 'B'
18         else:
19             print("Action: Move Left")
20             current_room = 'A'
21
22     if rooms['A'] == 'Clean' and rooms['B'] == 'Clean':
23         print("\nAll Rooms are Clean.")
24         break
25
26  print("Final Status:", rooms)

```

Terminal Output:

```

Status: Clean
Action: Move Right
Current Room: B
Status: Dirty
Action: Clean

All Rooms are Clean.
Final Status: {'A': 'Clean', 'B': 'Clean'}

```

Result

The Vacuum Cleaner problem was implemented successfully using a Simple Reflex Agent.

Conclusion

The Simple Reflex Agent successfully senses the environment, cleans dirty rooms, and moves between rooms until all rooms are clean. This demonstrates the basic working of an intelligent agent in Artificial Intelligence.